BMC Medicine

https://doi.org/10.1186/s12916-026-04748-5

Article in Press

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Received: 12 Dec 2025

Accepted: 23 Feb 2026

Published online: 11 April 2026

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**Cite this article as: Young, C., Garratt, M., Athukorala, T. *et al.* Maternal macronutrient intake and gestational diabetes: systematic review of prospective observational studies and dose–response meta-analyses. *BMC Med* (2026). https://doi.org/10.1186/s12916-026-04748-5**

# Maternal macronutrient intake and gestational diabetes: systematic review of prospective observational studies and dose-response meta-analyses

**Running title:** Maternal macronutrient intake on GDM

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**Maternal macronutrient intake and gestational diabetes: systematic review of prospective observational studies and dose-response meta-analyses**


**Abstract**

**BACKGROUND:** Gestational diabetes mellitus (GDM) is a common pregnancy complication associated with markedly elevated risk of type-2 diabetes in later life. Pregnancy or pre-pregnancy diet is a likely risk factor for GDM, with macronutrients – including proteins, fats, carbohydrates, and alcohol – being the primary dietary components to influence risk. In this study, we have conducted a systematic review and meta-analyses of pregnancy or pre-pregnancy dietary macronutrient intakes, including by source and type, on GDM occurrence to better elucidate the role of maternal diet on GDM.

**METHODS:** We ran an online search on 15 August 2025 in OVID Medline, EMBASE, and EBM Reviews, augmented with reference searching to identify prospective observational studies reporting pregnancy or pre-pregnancy dietary intakes and GDM incidence. Screening, data extraction, and risk of bias assessment using the Newcastle-Ottawa Scale were performed by at least two reviewers. Macronutrient exposures and GDM incidence were considered with random effects higher versus lower (HvL) and dose-response analyses. Certainty of evidence was then considered with GRADE to comment on maternal macronutrient intakes and GDM.

**RESULTS:** From 3,990 initial titles, we identified 25 prospective observational studies of 115,496 pregnant women reporting at least one macronutrient intake and GDM. There was moderate certainty evidence that animal-sourced protein (HvL: RR 1.62 (95% CI 1.20 to 2.18) $I^2$ 82%) and animal-sourced fat (HvL: RR 1.59 (95% CI 1.34 to 1.88) $I^2$ 0%) intakes were associated with higher GDM in a dose-response manner. Total protein (HvL: RR 1.63 (95% CI 1.22 to 2.18) $I^2$ 64%) and vegetable fat (HvL: RR 1.25 (95% CI 1.06 to 1.47) $I^2$ 0%)

showed positive associations with outcome in HvL but without dose response, while dietary fibre (HvL: RR 0.65 (95% CI 0.44 to 0.96) $I^2$ 89%) was the only macronutrient to indicate a protective effect against GDM with higher intakes in a dose-response manner, with moderate certainty evidence.

**CONCLUSIONS:** This study is the first to consider maternal intake of all macronutrients, including by source and type, on GDM occurrence. Our findings are largely in line with healthy dietary advice for the general population (promoting high fibre and low animal-sourced fat intakes), however higher protein intakes (both total and animal-sourced) may be a specific concern that is not currently considered in dietary guidance during pregnancy, requiring further research.

Trial registration: PROSPERO (CRD420251001515)

Keywords: Gestational diabetes; Risk; Diet; Macronutrients; Protein; Fat; Fibre; Systematic review; Dose-response; Meta-analysis



## Background

Pregnancy is a significant life event for many women[1], but it can be threatened by obstetric complications that endanger maternal and foetal health. The prevalence of gestational diabetes mellitus (GDM), the most common of these obstetric conditions, has been increasing steadily in recent decades, with an estimated 14% of all pregnancies worldwide being affected (1). GDM is defined as hyperglycaemia first diagnosed during pregnancy (2). It has a strong association with peripartum complications such as pre-eclampsia, foetal macrosomia, and birthing difficulties (3). Although the characteristic hyperglycaemia of GDM typically

[1] The terms ‘woman’, ‘mother’, and ‘maternal’ are used as this is the way the majority of those who are pregnant or having a baby will identify. We do acknowledge the complexity of gender and biological sex in the field of obstetrics and the subsequent limitations of using such terminology.

resolves post-delivery, women who have had GDM have a seven-fold increased risk of type-2 diabetes (4) and an elevated risk of cardiovascular disease and certain cancers (5, 6) in later life. Children born to GDM pregnancies similarly have an increased lifetime risk of developing cardiometabolic diseases (7).

Although there are many interacting genetic and lifestyle determinants of GDM risk, diet represents an important modifiable risk factor (8). When food is consumed, its constituent macronutrients are degraded into energy for bodily functions (9). Primary macronutrients in the body are proteins, fats, carbohydrates, and alcohol, although within these groupings they can vary by type and dietary source. Epidemiological studies have established a clear link between maternal diet quality and GDM onset (10). To date, however, no meta-analysis has systematically assessed the full range of macronutrient intakes on GDM incidence. Considering all macronutrients in combination is required to understand how variation in the intake of energy sources influences human health during pregnancy because changes in the intake of one macronutrient typically requires compensatory changes in another for energy balance to be maintained (11).



We have conducted a systematic review and dose-response meta-analysis to identify the role of maternal macronutrient intakes on GDM occurrence. As a modifiable risk factor, maternal macronutrient intake is an attractive clinical target for mitigating the risk of GDM and the aligned metabolic complications during and after pregnancy. Understanding gained from this work will be useful for identifying a range of optimal dietary intakes for maternal health.

## Methods

We followed Cochrane guidelines (12) for conducting systematic reviews and the preferred reporting items for systematic reviews and meta-analyses (PRISMA) (13). The protocol for this systematic review and meta-analysis was preregistered with PROSPERO

(CRD420251001515).

### *Eligibility criteria*

This systematic review addresses the research question, “What is the role of maternal macronutrient intake on GDM incidence?”. We considered the available data from peer-reviewed prospective observational studies where dietary data were captured pre or during pregnancy in adults, and GDM status was assessed later in pregnancy. Eligible dietary assessment included food frequency questionnaires (FFQ), food recall, food diary, or diet history. Prospective observational studies had to report macronutrient intakes and correlate intake with GDM occurrence to be eligible. We considered the overall macronutrient category (e.g., protein, fat, carbohydrate, and alcohol) as well as by type (e.g., sugar, starch, and fibre for carbohydrate) and source (e.g., animal or vegetable source for protein).



### *Literature search*

Potential search terms underwent sensitivity and specificity testing to ensure completeness before being finalised. This included testing variations of terms to maximise relevant results and checking search results to ensure key known papers were captured (see Additional file 1: Appendix A and Tables S1–S3). The search strategy was run in OVID Medline, EMBASE, and EBM Reviews from database inception to 15 August 2025. The online search strategy was supplemented with hand searches of reference lists to identify other potentially eligible publications. There were no date or language restrictions applied to searching. Title/abstract and full-text screening were undertaken independently by at least two authors. Disagreements between reviewers were resolved in discussion. Commercially available software (Covidence) was used to remove duplicates and aid screening (14).

### *Data extraction and risk of bias assessment*

Data were extracted from eligible studies using pretested forms (15) to both describe the general characteristics of the eligible studies and to enable statistical analyses. For descriptive characteristics, we collected author name, year published, macronutrient captured, country, study design, number of participants and cases, follow-up, dietary assessment type, pregnancy period of assessment, and adjustments in statistical models. For inferential statistics, we collected relative risk statistics and case numbers. The most adjusted effect size values were extracted to control for confounders in the diet-disease association we consider herein, as recommended in Cochrane guidelines (16). The Newcastle-Ottawa Scale (NOS) (17) was used to assess risk of bias for all eligible studies. The two important confounder variables considered with NOS were energy intake and body-mass index (BMI). Energy intake was calculated from the same dietary assessment technique as macronutrient intake and BMI was calculated per anthropometric measures at baseline. Data extraction and risk of bias assessment were undertaken by one reviewer with each extracted variable confirmed by at least one other author with conflicts resolved through discussion.



***Statistical analysis and assessment of evidence certainty***

Analyses were performed in Stata statistical software (version 19.5) using the metan, metabias, metarim, metainf, and metareg commands. We considered the relationship between macronutrient intake and GDM by comparing the highest intake quantile with the lowest intake quantile per macronutrient with random effects meta-analyses (18). We considered heterogeneity between the reported results of individual studies with the $I^2$ statistic (19). For studies where dietary assessment was undertaken at several time points, we first averaged the available risk association data with fixed-effects meta-analyses before pooling with other studies to avoid unit of analyses error.

Several sensitivity analyses were undertaken for each exposure-outcome analysis. Small study effects, such as potential publication bias, were assessed with Egger's test (20) and, if likely, the trim and fill method was applied to consider the impact of the small study effect (21) on the pooled result. Any analysis involving two or more studies was considered with an influence analysis where each study was removed from the pooled estimate one at a time to consider if any individual study substantially changed the reported result.

Categorical meta-regression analyses were then conducted on pooled data to consider underlying reasons for potential heterogeneity. Variables considered with categorical meta-regression were: when diet was assessed (e.g., pre-pregnancy and first, second, or third trimester), dietary assessment technique applied, decade of publishing, geographical region of the study, and the overall NOS score.

Dose-response relationships were considered with restricted cubic splines with three knots at the 10th, 50th, and 90th percentiles (22) of macronutrient intakes in a two-stage, random effects model (23). This process used all available quantile values for exposure (e.g., grams of fibre per day) and outcome (relative risk of GDM occurrence). After analysing each exposure-outcome pairing, we used Grading of Recommendations Assessment, Development and Evaluation (GRADE) protocols (24) to comment on the certainty of the evidence for each macronutrient in relation to GDM occurrence.



**Results**

After initial consideration of 3,990 titles, we identified 25 eligible prospective observational studies reporting on macronutrient exposure and GDM incidence as shown in Figure 1. Twenty-three were cohort studies and two were case-controls nested in cohort studies. The majority of studies were from Asia (56%) and North America (24%), followed by Europe (16%) and Oceania (4%). A total of 10,210 GDM cases from 115,496 pregnancies (8.8% GDM incidence) were captured. Study size ranged from 100 to 15,509 participants. Eight studies (4,272 cases of 73,888 participants) relied on participants self-reporting a GDM diagnosis (25-32), otherwise GDM was validated by accessing medical records or assessment within the study. Participants were followed for an average of 20.8 weeks and the most common form of dietary assessment was FFQ (72%), followed by 24-hour diet recall (16%), 3-day food diary (12%), and diet history (4%). All studies excluded participants with a prior history of GDM and/or diabetes and most studies (56%) excluded participants with a prior pregnancy or miscarriage.

<Figure 1>

Thirteen studies reported on protein intakes (25, 28-30, 33-41), 12 on carbohydrate intakes (25, 27, 32, 33, 36, 37, 39, 41-45), 11 on fat intakes (25, 26, 33, 37, 39, 41, 42, 46-49), and one on alcohol intakes (31). The majority of studies (84%) assessed preconception (25-28, 31, 32, 34, 38, 40, 43, 46, 49) and early pregnancy (29, 33, 34, 36-38, 40, 42, 44, 45, 47-49) macronutrient intake, typically through FFQ or interview at first ultrasound appointment (approximately 10–14 weeks gestation). Fewer studies (36%) assessed diet at mid-pregnancy (33-35, 39-41, 44, 45, 49) with two studies having additional dietary assessments in late-pregnancy (8%) (30, 45), occurring after GDM diagnosis (i.e., late in the third trimester). Descriptive details of each eligible study are shown in Additional file 1: Table S4.



Analyses of all macronutrients, including by type and source, on GDM occurrence is shown in Table 1. Dose response associations are shown in Figures 2–5. Summary forest plots, meta-regression analyses per exposure, and certainty of evidence and GRADE tables are in Additional file 1: Figures S1(A–O) and Table S5–S6.

<Table 1>

***Protein intakes and GDM***

Higher intakes of both total and animal-sourced protein were associated with a greater incidence of GDM when compared with lower intakes, indicating 68 (95% CI 24 to 99) and 65 (95% CI 21 to 124) more cases for higher consumers per 1,000 pregnancies. High intakes of vegetable-sourced protein did not have a significant association with GDM incidence (RR 0.90, 95% CI 0.73 to 1.10). Only animal-sourced protein intakes (RR 1.62, 95% CI 1.20 to 2.18) exhibited a dose-response relationship with GDM, with higher intakes associated with greater rates of GDM. The certainty of evidence for animal-sourced protein intakes on GDM was low, being downgraded once for unexplained heterogeneity. No meta-regression analyses indicated an underlying factor influencing the associations between protein intakes and GDM.

<Figure 2>

***Carbohydrate intakes and GDM***

Higher intake of fibre was associated with a lower incidence of GDM when compared with lower intake, accounting for 31 (95% CI 4 to 50) fewer cases per 1,000 pregnancies. Both total carbohydrates (RR 0.78, 95% CI 0.54 to 1.15) and sugar (RR 0.83, 95% CI 0.56 to 1.23) did not have a significant association with GDM incidence. Dose-response analysis indicated that higher fibre intake showed greater protection against GDM in an increasing fashion up until the highest recorded intakes of 24.9 g per day. No meta-regression analyses indicated an underlying factor influencing the associations between carbohydrate intakes and GDM. Certainty of evidence was very low for total carbohydrates, moderate for fibre, and not assessed for sugar as only one relevant prospective observational study was identified.

<Figure 3>

***Fat intakes and GDM***

Higher intakes of animal and vegetable fats were associated with a greater incidence of GDM when compared with lower intakes, accounting for 40 (95% CI 23 to 60), and 17 (95% CI 4 to 32) more cases per 1,000 pregnancies. Higher intakes of total fats did not have a significant association with GDM incidence (RR 1.33, 95% CI 0.99 to 1.77). Only animal-sourced fats exhibited a dose-response relationship with GDM, whereby increasing intake was associated with higher GDM risk. There was moderate certainty evidence in this association. Meta-regression of total NOS scores (<5 versus >6) identified one high risk study that may over-estimate the association of total fat (seven low risk studies pooled RR 1.23, 95% CI 0.88 to 1.71).

<Figure 4>

***Alcohol intake and GDM***

Only one study reporting alcohol intake in the maternal diet and GDM incidence was identified, with the data inconclusive of any relationship.

<Table 1>

**Discussion**

We have examined the relationship between maternal macronutrient intake and GDM with a systematic review and meta-analysis of prospective observational studies. We have identified that dietary profiles with higher intakes of total and animal-sourced protein or vegetable and animal-sourced fats are associated with higher incidence of GDM when compared with lower intakes. Conversely, higher intake of dietary fibre are likely protective against GDM incidence. These findings were present using maximally adjusted models, which routinely took into account all energy intake indicating possible direct, rather than bodyweight-mediated, associations.

Considering the full range of macronutrients provides a holistic picture of how dietary components collectively influence GDM risk. This is essential for refining nutritional guidelines for pregnancy, where recommendations for consumption of specific food sources would typically cause changes in the intake of multiple macronutrients simultaneously. According to the results here, for intake of multiple macronutrients pre- or during pregnancy, animal sources of protein and fat are associated with GDM, while total carbohydrate intake does not influence GDM risk, and high fibre intake reduces risk. Thus, tailored substitution of animal sources of protein and fat for high-fibre carbohydrate would be expected to reduce GDM incidence in some individuals, particularly those initially consuming high amounts of red meat and low amounts of fibre.

We are the first to consider the role of all macronutrients in the maternal diet on GDM, rather than considering a single macronutrient or those from specific sources only. Our findings linking animal-sourced protein and fat intakes with increased GDM risk have been observed with meta-analyses previously (50, 51), although our findings derived from more data indicate less inflated associations. Similarly, previous meta-analyses for total fat (52, 53) indicate greater GDM incidence with higher intakes, however with our more recent search identifying more studies, we did not observe a statistically significant effect of total fat intake on incidence rates.



These data on protein and fat together indicate that lower intakes of animal products (such as meat and dairy) may reduce the risk of GDM. However, we recognise that animal-sourced protein and fat almost always exist in the same foods, so any relative adjusting for the other macronutrient in the source papers may be subject to colinearity issues. It may be that the strong effect of animal protein is increased by excess animal fat intake, or vice versa, with future research with other study designs such as animal studies on controlled diets potentially useful to better understand the associations we observed here. Additionally, animal studies would provide insight to the concurrent effects of macronutrients on entire body systems, as food is rarely consumed as single macronutrients so observing its systemic effects is important. For example, a recent study testing the effects of macronutrients during pregnancy reported that glucose tolerance in pregnant mice is progressively impaired with increasing levels of dietary protein intake, but is unaffected by the level of dietary fat content (54). This supports a direct role of protein in causing impaired glucose tolerance during pregnancy, which may contribute to increased GDM in women consuming high protein diets.

Pregnancy physiology is complex and, while the complete mechanisms underlying the effects of macronutrients are unclear, some pathways are suggested to be implicated in GDM pathogenesis. Animal-sourced proteins have a higher abundance of branch-chained amino acids (BCAAs – leucine, isoleucine, valine) and methionine which have been shown in human studies to be associated with GDM (28, 55). In non-pregnant studies, excess circulating BCAAs can result in a state of insulin resistance via activation of the mammalian target of rapamycin complex 1 (mTORC1) (56). During pregnancy, placenta-derived hormones such as human chorionic gonadotropin and human placental lactogens promote a mild state of insulin resistance to help promote nutrition provisioning to the foetus. The maternal pancreas adapts to compensate for this, increasing insulin secretion and maintaining glucose homeostasis. However, in situations of high protein intake (i.e., excess amino acids, causing a state of nutrient overload), the additive effects of amino acids and placental hormones may more strongly impair insulin resistance and contribute to the pathological hyperglycaemic state of GDM (57, 58).

Dietary fats are also an important part of maternal nutrition as they support pregnancy metabolism and foetal development. Animal-sourced fats are more abundant in saturated fatty acids (SFAs) compared with vegetable-sourced fats, and SFAs are strongly associated with insulin resistance and diabetes pathogenesis (59). Higher intakes of SFAs can lead to excess fatty acids in circulation which acculumate in tissues such as liver and muscles and disrupt insulin signalling (60). SFAs also have a direct action on inflammatory pathways such as Toll-like receptor 4 (TLR-4) in the liver (61) which contribute to a chronic inflammatory state characteristic of GDM. The negative effects of SFAs on insulin signalling may contribute to the dose-dependent relationship between animal fats and GDM observed in this study, although there are likely other factors given that our HvL analysis did not identify a significant effect of SFA itself on GDM incidence. Conversely, intake of vegetable-sourced nutrients such as dietary fibre promotes glucose homeostasis. Dietary fibre, considered an essential nutrient (62), lowers postprandial hyperglycaemia and glycaemic variability and, when consumed regularly, can improve long-term glycaemic control as measured by glycated haemoglobin (63-65). This suggests direct pathways for glycaemic control and may explain why we observed a protective dose-response effect in this study.



Broadly, our findings are in line with global dietary recommendations of GDM prevention and management (66) as well as with guidelines for the population (67-70) and those with diabetes (71). Key messages across recommendations are the base eating around minimally processed plant foods such as whole grains, vegetables, whole fruit, legumes, nuts, seeds, and non-hydrogenated non-tropical vegetable oils, while minimising the consumption of red and processed meats, sugar-sweetened beverages, and refined grain foods. Diets based on these food groups would provide the appropriate maconutrient composition identified in this research (i.e., high fibre, low in animal-sourced protein and fat) as beneficial for GDM risk reduction. Diets with extreme intakes of any maconutrient are generally contraindicated for pregnancy and overall health (68, 70). While these dietary messages exist, and our analyses are broadly supportive of these recommendations in the context of maternal diet and GDM risk, it is worth recognising that very few people follow this guidance. Animal-sourced fat intakes are normally above recommendations, while dietary fibre is a shortfall nutrient (72) requiring an increase in intakes. Our findings indicate that protein intake, while extremely important for foetal development, might need to be moderated if women are already consuming high levels of protein at the upper levels of normal dietary guidelines (i.e., greater than 20% of total energy).

Strengths of this study are its novel, comprehensive approach to considering all macronutrients, and the use of dose-response testing and robust sensitivity analyses to check each exposure-outcome pairing. The health outcome considered, GDM, is a globally relevant concern increasing in incidence, particularly within Asian and European populations (73). Primary limitations are the relatively acute measure of dietary intake and the reliance on self-reported dietary intakes. Although most studies used the International Association of the Diabetes and Pregnancy Study Groups (IADPSG)/World Health Organization (WHO) diagnostic criteria of GDM (fasting plasma glucose (PG) $\geq$ 5.1 mmol/L and/or 1-hour PG $\geq$ 10.0 mmol/L and/or 2-hour PG $\geq$ 8.5 mmol/L), some still used pre-existing local criteria (e.g., fasting PG $\geq$ 5.6 mmol/L and/or 2-hour PG $\geq$ 7.8 mmol/L) which may underestimate the prevalence of GDM (74) and affect these results. Although we found no association between geographical region of publication and GDM incidence from meta-regression analyses, we identified no relevant studies from Africa, Middle East, or South America, potentially reducing global representativeness of our pooled results. Countries known to have higher rates of GDM (e.g., Qatar, India, Vietnam, Finland, Bangladesh (1)) were also not represented in this dataset. We also identified fewer studies for some macronutrient types, such as for sugars intake. This is an important areas for future research, given that other study designs have linked excess sugar intake with GDM (75).

**Conclusions**

We have conducted the first study to systematically assess the full range of macronutrient intakes, including by type and source, with GDM incidence. Our findings indicate that maternal diets – such as those high in animal-sourced protein and fat and low in dietary fibre – are of elevated risk for GDM, providing evidence for potential dietary interventions and practical clinical advice to patients. Our findings are largely in line with healthy dietary advice for the general population, however higher protein intakes (both total and animal-sourced) may be a specific concern for this patient population, requiring further research.

**Abbreviations**

BCAAs: Branch-chained amino acids

BMI: Body mass index

FFQ: Food frequency questionnaires

GDM: Gestational diabetes mellitus

GRADE: Grading of Recommendations Assessment, Development and Evaluation

IADPSG: International Association of the Diabetes and Pregnancy Study Groups

mTORC1: Mammalian target of rapamycin complex 1

NOS: Newcastle-Ottawa Scale

PG: Plasma glucose

PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses

SFAs: Saturated fatty acids

TLR-4: Toll-like receptor 4

**Declarations**

**Ethics approval and consent to participate:** Not applicable.

**Consent for publication:** Not applicable.

**Availability of data and materials:** The datasets supporting the conclusions of this article are included within the article and its additional files. Raw data that support the findings of this study are available from the corresponding author, upon reasonable request.

**Competing interests:** The authors declare that they have no competing interests.

**Funding:** This research received no external funding. CDY was supported by a University of Otago Doctoral Scholarship. ANR was supported by the Heart Foundation-Hynds Senior Research Fellowship.



**Authors' contributions:** CDY: designed the research, conducted searches, screening, data extraction, conducted the analyses, interpreted the results, and led manuscript writing. MGG: designed the research, interpreted the results, reviewed manuscript. TA: screening, data extraction, reviewed manuscript. TP, AP, SC, MP: reviewed manuscript. ANR: designed and oversaw the research, conducted screening, interpreted the results, and supported manuscript writing. All authors read and approved the final manuscript.

**Acknowledgements:** We thank Lynne Knapp (University of Otago) for supporting the development of search terms.

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**List of Tables**



**List of Figures**

(E) Polyunsaturated fat intake and GDM occurrence. 1,507 cases over 18,472 person-years. (F) Monounsaturated fat intake and GDM occurrence. 1,507 cases over 18,472 person-years.

## Additional Files

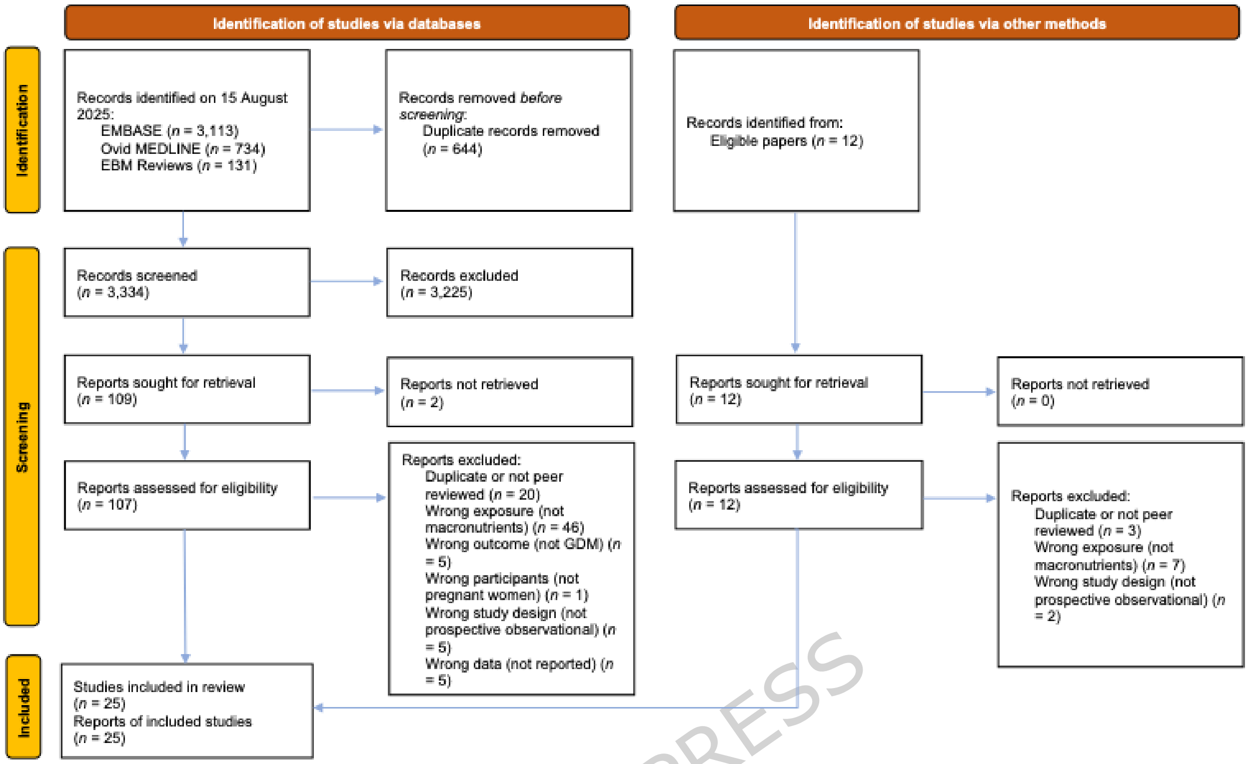
**Additional file 1:** Appendix A Supporting information relating to the methodology. Table S1 Online search term development. Table S2 Papers used for specificity testing. Table S3 Results from online literature search. Table S4 Descriptive details of eligible prospective observational studies. Figures S1(A–O) Higher versus lower analyses of macronutrient intake and GDM occurrence. Table S5 Meta-regression analyses per exposure. Table S6 GRADE Tables.

**Table 1.**

| **Dietary exposure** | **Number of studies** | ***N* pregnant adults (cases)** | **Relative risk (95% CI)** | **Absolute risk (95% CI)** | **$I^2$** | **Dose response?** | **GRADE** |
|---|---|---|---|---|---|---|---|
| **Protein** | | | | | | | |
| Total protein | 9 | 30,637 (3,308) | 1.63 (1.22, 2.18) | 68 more per 1,000 (from 24 more to 99 more) | 64.1% | No | Very low |
| Animal protein | 8 | 29,742 (3,131) | 1.62 (1.20, 2.18) | 65 more per 1,000 (from 21 more to 124 more) | 82.2% | Yes | Moderate |
| Vegetable protein | 8 | 29,742 (3,131) | 0.90 (0.73, 1.10) | 11 fewer per 1,000 (from 28 fewer to 11 more) | 59.9% | No | Very low |
| **Carbohydrate** | | | | | | | |
| Total carbohydrate | 7 | 27,619 (1,959) | 0.78 (0.54, 1.15) | 16 fewer per 1,000 (from 28 fewer to 11 more) | 78.0% | No | Very low |
| Fibre | 8 | 38,142 (3,386) | 0.65 (0.44, 0.96) | 31 fewer per 1,000 (from 50 fewer to 4 fewer) | 88.7% | Yes | Moderate |
| Sugar | 1 | 6,263 (285) | 0.83 (0.56, 1.23) | 8 fewer per 1,000 (from 20 fewer to 10 more) | - | - | - |
| **Fat** | | | | | | | |
| Total fat | 9 | 22,474 (2,519) | 1.33 (0.99, 1.77) | 37 more per 1,000 (from 1 fewer to 86 more) | 83.7% | No | Very low |
| Animal fat | 4 | 24,006 (1,638) | 1.59 (1.34, 1.88) | 40 more per 1,000 (from 23 more to 60 more) | 0.0% | Yes | Moderate |
| Vegetable fat | 4 | 24,006 (1,638) | 1.25 (1.06, 1.47) | 17 more per 1,000 (from 4 more to 32 more) | 0.0% | No | Low |
| SFA | 5 | 27,313 (2,291) | 1.05 (0.91, | 4 more per 1,000 (from 8 | 0.0% | No | Very low |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | 1.21) | fewer to 18 more) | | | |
| PUFA | 4 | 24,976 (2,034) | 0.90 (0.64, 1.28) | 8 fewer per 1,000 (from 29 fewer to 23 more) | 82.9% | No | Very low |
| MUFA | 4 | 24,976 (2,034) | 1.18 (0.86, 1.62) | 15 more per 1,000 (from 11 fewer to 50 more) | 65.6% | No | Very low |
| Trans fat | 2 | 17,200 (1,504) | 1.05 (0.89, 1.24) | 4 more per 1,000 (from 10 fewer to 21 more) | 0.0% | No | Low |
| Unsaturated fat | 1 | 2,337 (257) | 1.00 (0.71, 1.40) | 0 fewer per 1,000 (from 32 fewer to 44 more) | - | - | - |
| n-6:n-3 PUFA ratio | 1 | 100 (22) | 4.29 (1.3, 14.12) | 724 more per 1,000 (from 66 more to 1,000 more) | - | - | - |
| **Alcohol** | | | | | | | |
| Total alcohol | 1 | 14,284 (1,209) | 0.93 (0.84, 1.05) | 5 fewer per 1,000 (from 12 fewer to 4 more) | - | - | - |

**Figure 1.**


Identification of studies via databases
Identification of studies via other methods
Identification
Screening
Included
Records identified on 15 August 2025:
EMBASE (n = 3,113)
Ovid MEDLINE (n = 734)
EBM Reviews (n = 131)
Records removed before screening:
Duplicate records removed (n = 644)
Records identified from:
Eligible papers (n = 12)
Records screened (n = 3,334)
Records excluded (n = 3,225)
Reports sought for retrieval (n = 109)
Reports not retrieved (n = 2)
Reports sought for retrieval (n = 12)
Reports not retrieved (n = 0)
Reports assessed for eligibility (n = 107)
Reports excluded:
Duplicate or not peer reviewed (n = 20)
Wrong exposure (not macronutrients) (n = 46)
Wrong outcome (not GDM) (n = 5)
Wrong participants (not pregnant women) (n = 1)
Wrong study design (not prospective observational) (n = 5)
Wrong data (not reported) (n = 5)
Reports assessed for eligibility (n = 12)
Reports excluded:
Duplicate or not peer reviewed (n = 3)
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Wrong study design (not prospective observational) (n = 2)
Studies included in review (n = 25)
Reports of included studies (n = 25)

Figure 2.

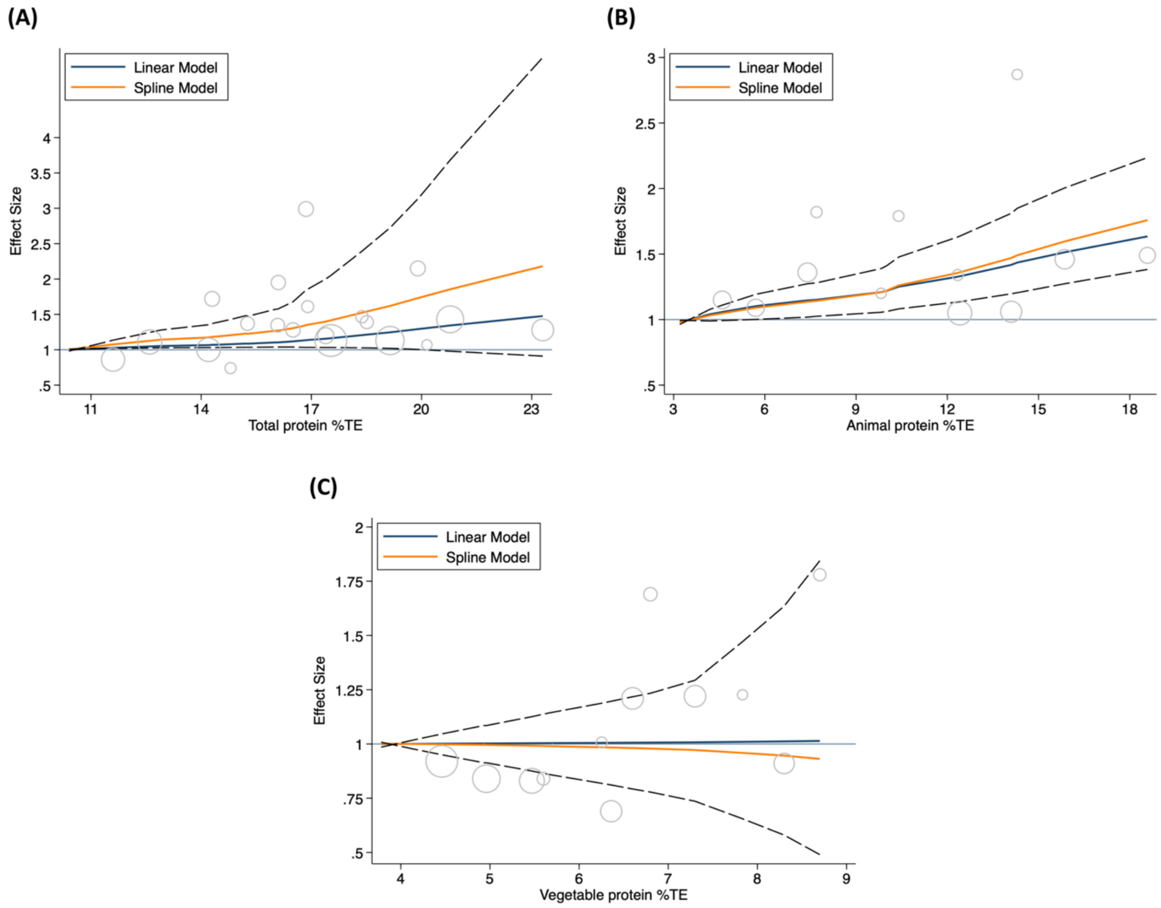





Figure 3.

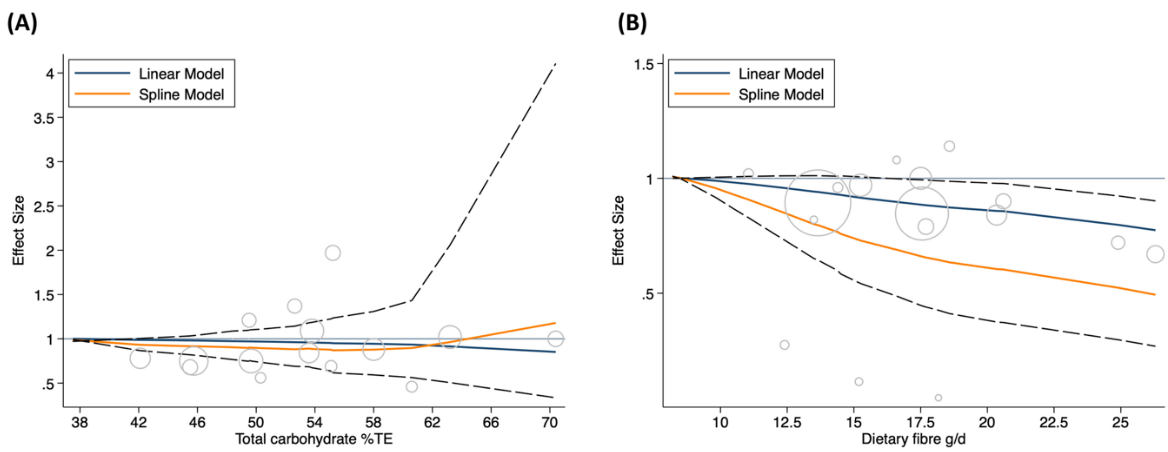
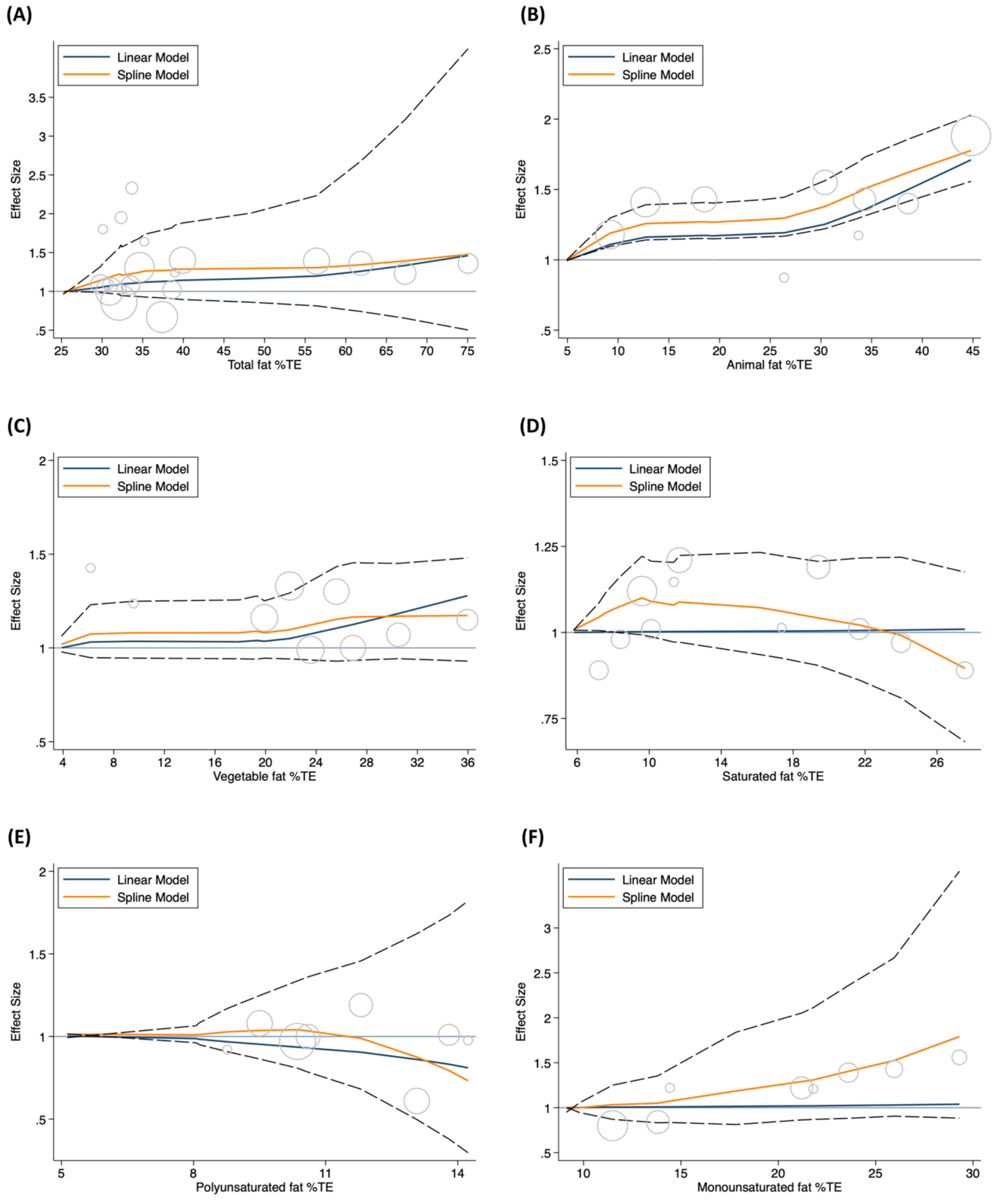


**Figure 4.**


(A)
Linear Model
Spline Model
Effect Size
Total fat %TE
(B)
Linear Model
Spline Model
Effect Size
Animal fat %TE
(C)
Linear Model
Spline Model
Effect Size
Vegetable fat %TE
(D)
Linear Model
Spline Model
Effect Size
Saturated fat %TE
(E)
Linear Model
Spline Model
Effect Size
Polyunsaturated fat %TE
(F)
Linear Model
Spline Model
Effect Size
Monounsaturated fat %TE